

Professional Practices

Lecture 2

What is Bioethics?

- Bioethics is the interdisciplinary study of ethical, social, and legal issues arising from advances in biomedicine, biotechnology, and the life sciences. It addresses moral dilemmas in healthcare ([medical ethics](#)), research ([research ethics](#)), and environmental policy. Key principles include [autonomy](#), [beneficence](#), [non-maleficence](#), and [justice](#).

Key Aspects of Bioethics

Key Aspects of Bioethics:

- **Definition:** A field combining biology (life) and ethics (moral principles) to address questions about what is "good" or "bad" in science and health.
- **Scope:** Covers topics such as end-of-life care, genetic engineering, reproductive technology, organ transplantation, and the allocation of limited healthcare resources.
- **Core Principles:** The "four principles" approach developed by Beauchamp and Childress is foundational:
 - **Autonomy:** Respecting a patient's right to self-determination.
 - **Beneficence:** Acting in the best interest of the patient.
 - **Non-maleficence:** "Do no harm".
 - **Justice:** Ensuring fairness in the distribution of benefits and risks.
- **Purpose:** To guide clinical decisions, develop policy, and manage ethical conflicts in medicine and research. 

Bioethics is critical for addressing complex, value-based decisions in a rapidly advancing technological world. 

Introduction To Bioethics

We know Bio = life

Ethos = behavior/character

....but what is that really?

Ethics is quite complicated.

How do you define Ethics?

Many varied definitions...

What does it mean to you?

- ***Ethics*** - sometimes used as synonyms for “**morally** correct” or justified - set of justified moral principles of obligation, rights, and ideals
- ***Ethics*** - can be particular **beliefs** or attitudes concerning morality
- ***Ethics*** - area of **study or inquiry** – an activity of understanding moral values, resolving moral issues, and justifying moral judgments

A

- In one sense, the word **ethics** and its grammatical variants are **synonyms for “morally correct” or justified**. In this usage, engineering ethics amounts to the **set of justified moral Principles of obligation, rights, and ideals** that ought to be endorsed, as they apply generally to engineering in particular, by those engaged in engineering. Clarifying such principles and applying them to concrete situations is the central goal of engineering ethics as an area of study.

B

- In a second sense the word **ethics** refers to the **particular beliefs or attitudes concerning morality** that are endorsed by specific groups or individuals. Using this sense, engineering ethics consists of the requirements specified in the currently accepted engineering codes of ethics.
- Alternatively, engineering ethics may refer to the actual conduct of individuals who are engineers.

C

- In the sense we will use most **ethics** refers to an **area of study or inquiry – an activity of understanding moral values, resolving moral issues, and justifying moral judgments.**

Values, Morals, Ethics

- **Values** signify what is important and worthwhile. They serve as a basis for moral codes and ethical reflection.
- **Morals** are codes of conduct governing behavior. They are values put into practice as actions.
- **Ethics** provide a systematic, rational way to work through dilemmas and to determine the best course of action in the face of conflicting choices.

Ethics is related to morals and values.

What are moral values?

What is morality?

- **Dictionary** says – concerns right and wrong, good and bad, the rules that ought to be followed.
- Latin MOS = custom
- **Morality** – about reasons centered in respect for other people as well as ourselves, reasons that involve their good as well as our own.
- Are everyone's morals the same? (Guantanamo)
- Different cultures? (Persian King) Different times? (slavery)

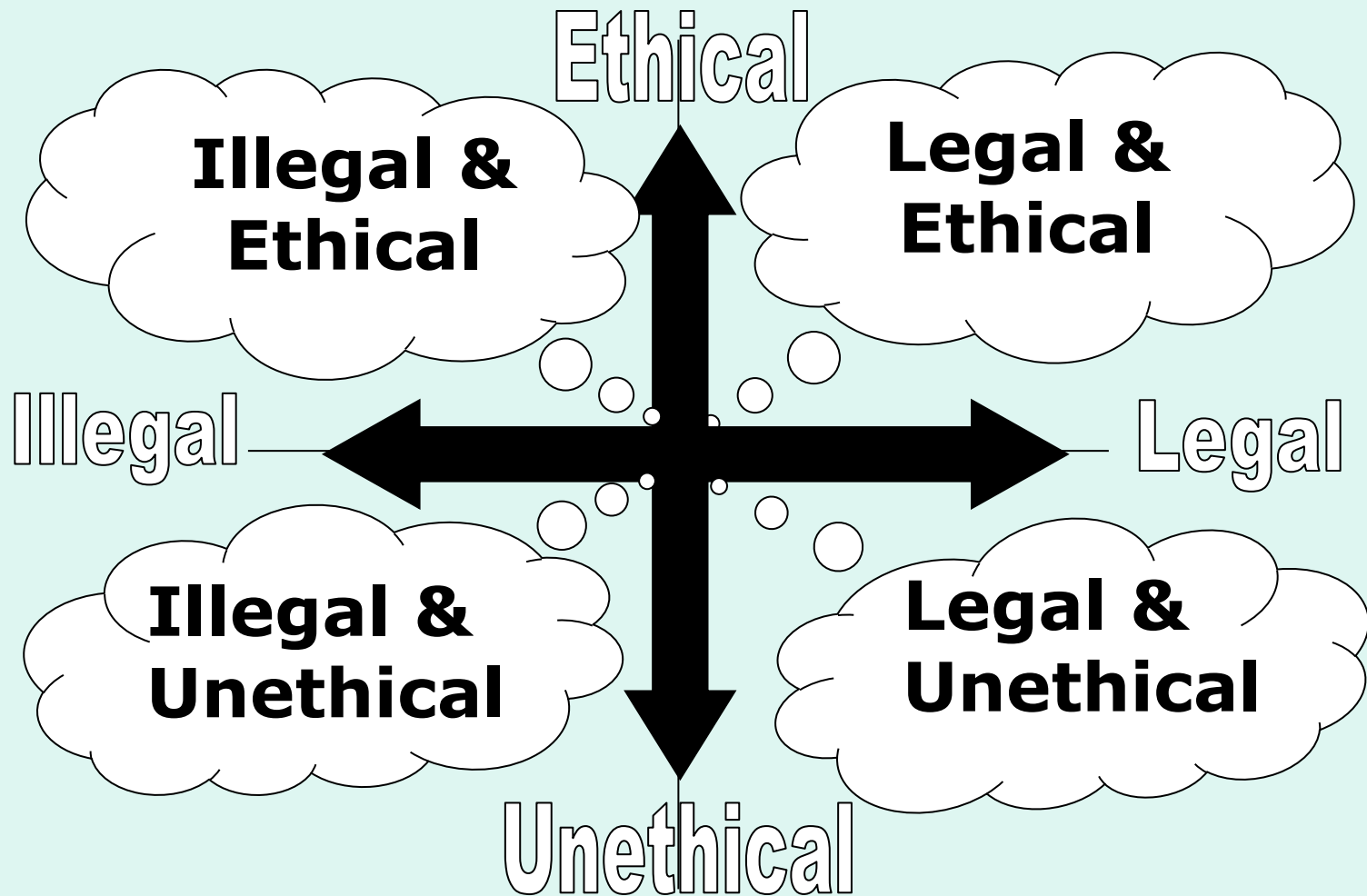
Continue...

- Dictionary definition is not necessarily very helpful in putting into practice.
- There are a number of theories of the development of ethics and morals. We will not have time to develop those here, but a course (or independent reading) in ethics or engineering ethics in particular would likely provide that.
- Working definition for our purpose could be: Morality – about reasons centered in respect for other people as well as ourselves, reasons that involve their good as well as our own.
 - fair and just
 - respect for rights of others
 - keeping promises
 - avoiding cheating and dishonesty

Developing a Well-Reasoned Response to a Moral Dilemma

- Moral dilemmas are situations in which two or more moral obligations, rights, or ideals come into conflict with one another.
- Figure out:
Who are the stakeholders.
What are the options?
What are your justifications? Are they weak or strong reasons?
- How does one decide whether a response is well-reasoned? What criteria apply? How can we reliably judge?

How are ethics and law related?



Roles of Codes of Conduct

- Shared Standards
- Positive Support to Act Ethically
- Guidance Concerning Obligations
- Motivation to be ethical
- Education to be ethical
- Deterrence to being unethical
- Personal Discipline
- Professional Image

Roles of Codes of Conduct

- **Shared Standards**

The great diversity of moral views makes it essential that professions establish explicit standards. Shared standards throughout the profession has values, even if not idea.

- **Positive Support to Act Ethically**

Codes give positive support to those seeking to act ethically. “I am bound by the code of ethics of my profession to ...”

Roles of Codes of Conduct

- **Guidance Concerning Obligations**

Give helpful guidance concerning the main obligations of engineers. More specific directions may be in supplemental materials or can be developed by discussion based on fundamental guidance. Give forum to discuss.

- **Motivation to be ethical**

Reflections profession's positive statement of intent. Show that you have support of others for following positive ethical practices.

Roles of Codes of Conduct

- **Education to be ethical**

Good tool for classroom and continuing education. Case studies published every issue of NSPE* Newsletter for example. These examples and discussion help develop common understanding and experience.

*NSPE: National Society of Professional Engineers

Roles of Codes of Conduct

- **Personal Discipline and Deterrence(avoidance) to being unethical**

Can serve as the formal basis for investigation of unethical conduct. Professional societies do suspend members based on violations.

Roles of Codes of Conduct

- **Professional Image**

Positive image to the public of a committed professional. Represents self-regulation of the profession that helps reduce other regulation.

Need to acknowledge that code can be abused and does have limitations. For example, people may want to protect public image (of themselves or their company) when wrong has occurred or want to protect the status quo when progress is needed. Codes are no substitute for individuals grappling with concrete dilemmas.

**Basic Principles to be taken into consideration
(can be at odds with each other)**

- Non-maleficence (do no harm)
- Beneficence (do good)
- Justice (be fair)
- Truthfulness
- Confidentiality (doctor/patient)
- Autonomy (capacity of a rational individual to make an informed, un-coerced decision)

Non-maleficence (do no harm)

Non-maleficence, derived from the Latin phrase *primum non nocere* ("first, do no harm"), is a core principle of bioethics obligating healthcare providers to not intentionally cause harm, inflict pain, or incapacitate patients. It requires weighing treatment benefits against burdens, ensuring that risks are minimized or justified. 

Key aspects and applications include:

- **Definition:** Refraining from actions that harm patients, such as negligence, causing unnecessary pain, or killing.
- **Examples:** Stopping ineffective medication, avoiding negligent actions like improper treatment, and ensuring staff competence, as noted by the [IntelyCare website](#) and [Dr.Oracle](#).
- **Clinical Application:** It is central to end-of-life decisions regarding withholding/withdrawing treatment and managing pain, as explained on the [National Institutes of Health \(NIH\) | \(.gov\) website](#).
- **Balancing Act:** Often balanced with beneficence (doing good), meaning a treatment with potential harm (e.g., chemotherapy side effects) may be permissible if the benefit outweighs the risk.
- **Origin:** Rooted in the Hippocratic Oath, which promises to "abstain from doing harm". 

Beneficence (do good)

Beneficence is the ethical principle of acting with the intent to benefit others, promoting their well-being, and actively taking steps to improve their situation. It goes beyond merely avoiding harm (non-maleficence) by requiring proactive, often charitable, actions that serve a patient's or community's best interests. 

Key aspects and examples of beneficence include:

- **Healthcare & Nursing:** Examples include administering pain medication, promoting healing, educating patients, and acting in their best interest. It is often balanced with patient autonomy, ensuring "good" aligns with the patient's own values.
- **Proactive Care:** It requires positive steps, such as performing a beneficial, though risky, procedure to save a life, or providing preventive care to avoid future illness.
- **Professional Duty:** Beyond medicine, it is a guiding principle in education (e.g., tutoring a struggling student) and social work (e.g., advocating for vulnerable populations).
- **Broader Application:** It includes corporate social responsibility, such as eco-friendly practices, and the development of beneficial technology. 

Justice (be fair)

Justice is the consistent, impartial, and ethical practice of ensuring individuals receive their due, treating equals equally, and upholding rights regardless of status. It functions as a fair, principled, and often legal framework for resolving disputes, balancing competing interests, and fostering accountability. 

Key aspects of justice as fairness include:

- **Fundamental Fairness:** Actively providing a fair opportunity to be heard and treating all parties with dignity.
- **Procedural Integrity:** Relying on established rules, such as the rule of law, to ensure outcomes are unbiased.
- **Distributive Justice:** Ensuring a fair distribution of benefits and burdens within society.
- **Equality and Rights:** Upholding rights regardless of race, sex, or background, and protecting against exploitation. 

Truthfulness

Truthfulness in bioethics is the fundamental duty of healthcare professionals to provide accurate, honest information regarding diagnosis, prognosis, and treatment risks to uphold patient autonomy, foster trust, and enable informed decision-making. While modern standards demand full disclosure, exceptions exist for "therapeutic privilege" (preventing severe harm) or when patients express a desire not to know. 

Key Aspects of Truth-Telling in Healthcare

- **Autonomy and Consent:** Truthfulness is necessary for patients to exercise their autonomy and make informed decisions about their own care.
- **Building Trust:** Openness and honesty are essential for establishing and maintaining trust in the patient-provider relationship.
- **Shifting Standards:** Historically, paternalistic approaches allowed for withholding information, but modern ethics favor transparent communication.
- **Cultural Context:** In some cultures, families may request to withhold information from the patient, creating a conflict with Western patient-centered autonomy models.
- **Exceptions:** Therapeutic privilege is a limited exception where a doctor may withhold information if they believe disclosure would cause direct, serious harm (e.g., severe psychological distress). 

Confidentiality (doctor/patient)

Doctor-patient confidentiality is a core bioethical principle requiring clinicians to protect patient information, fostering trust and respecting autonomy. While generally obligated to keep information secret, physicians may breach confidentiality to prevent imminent, serious harm to the patient or others, or when required by law. 

Key aspects of confidentiality in bioethics include:

- **Ethical Basis:** Rooted in the Hippocratic Oath, it is vital for ensuring patients feel safe sharing sensitive, necessary information for their care.
- **Scope:** Covers all information gathered during treatment, including verbal, written, and electronic medical records.
- **Exceptions (Breaching Confidentiality):**
 - **Imminent Harm:** To prevent serious, unavoidable danger to the patient or identifiable third parties (e.g., infectious disease, violence).
 - **Legal Requirements:** Mandatory reporting of issues like gunshot wounds, child/elderly abuse, or certain communicable diseases.
 - **Public Interest:** In rare cases where public health is at stake, such as a major epidemic.

Confidentiality (doctor/patient)

- **Implementation & Challenges:**

- **Informed Consent:** Patients should know how their data is used and shared for care.
- **Modern Challenges:** The rise of electronic medical records and team-based care can lead to accidental disclosures, necessitating careful management.
- **Posthumous Privacy:** Confidentiality duties often extend beyond the death of the patient. 

Conclusion

Confidentiality is not absolute but a fundamental obligation that balances patient rights with safety, requiring careful ethical judgment when considering disclosure to third parties. 

Autonomy (capacity of a rational individual to make an informed, un-coerced decision)

Autonomy in bioethics is the principle recognizing a patient's right to self-determination, allowing competent individuals to make informed, voluntary decisions about their own medical care, including accepting or refusing treatment. It is one of the four core pillars of medical ethics, ensuring respect for personal values, bodily integrity, and informed consent. 

Key Aspects of Autonomy:

- **Informed Consent:** Patients must receive sufficient, understandable information regarding risks, benefits, and alternatives to make a voluntary decision.
- **Capacity:** The patient must possess the mental competence to understand the information and consequences of their choice.
- **Freedom from Coercion:** Decisions must be free from controlling influences by others (doctors, family).
- **Respect for Refusal:** It empowers patients to refuse life-sustaining treatment, a key component of modern medical law. 

How to solve an ethical problem

1. **Identify the Problem:** Clearly define the issue and determine if it is truly an ethical dilemma or a legal/policy violation.
2. **Gather Information:** Collect all relevant facts, including the history of the situation and the perspectives of those involved.
3. **Identify Stakeholders:** Determine who will be affected by the decision.
4. **Evaluate Options:** Analyze potential actions using ethical principles (e.g., maximizing good, minimizing harm, respecting rights).
5. **Apply Ethical Tests:** Use tests such as the "newspaper test" (would you be comfortable if this was published?), the "reversibility test" (would you agree if you were on the other side?), or checking against codes of conduct.
6. **Make a Decision and Act:** Implement the chosen, defensible solution.
7. **Reflect:** Evaluate the outcome to learn for future situations. 

Key Strategies for Ethical Decision-Making

- **Consult Experts:** Seek advice from mentors, ethics committees, or professional organizations.
- **Brainstorm Alternatives:** Look for "third way" solutions that satisfy conflicting values rather than a simple right/wrong choice.
- **Use Emotional Intelligence:** Manage emotions and use empathy to understand different perspectives, particularly in high-stakes workplace situations.
- **Lead by Example:** Model ethical behavior to foster a culture of integrity. 

How to solve an ethical problem based on consensus

- Consensus?
 - Based on reason
 - Genuine debate
 - Takes ethical traditions into account
 - Open to criticism, refutation and the possibility of change and exchange of ideas
 - It is not majority opinion as it often needs to protect the minority
 - Takes time
 - NOT SOLVED IN ISOLATION!

To do

- 5 examples of ethics/legal axis.
- Point out an ethical issue and proposed a solution.